

Key Data Processing Parameters in DeepHalo

Module / Section	Parameter	Value / Setting (default)	Description
FeatureFilter	RT_min	"	Minimum retention time (in seconds) to keep MS data. Leave empty (") to keep all data (no RT filter). Suggestion: Exclude the system void volume and non-retained components (to) to minimize false positives caused by severe co-elution.
	RT_max	"	Maximum retention time (in seconds) to keep MS data. Leave empty (") to keep all data (no RT filter). Suggestion: Exclude late-eluting regions under extreme organic phase conditions (e.g., >95% B) to minimize false positives caused by co-elution of strongly retained matrix interferences
	H_score_threshold	0.4	Confidence threshold for halogenated compound detection (Range: 0-1, higher is stricter).
	Anomaly_detection_threshold	7.02e ⁻⁰⁵	Threshold for isotope peak intensity reconstruction error. Use 1.0 to disable filtering.
FeatureMapProcessor	min_num_of_masstraces	3	Minimum number of isotope peaks a feature must have.
	min_feature_int	5000	Minimum summed intensity of a candidate feature.
	min_scan_number	3	Minimum number of consecutive scans a feature must span.
	use_mass_difference	"	Enables mass difference correction for isotope patterns. Set to 'none' to disable.
Dereplication	error_ppm	10	Precursor mass tolerance (ppm) for database matching.
	Inty_cosine_score	0.96	Minimum cosine similarity threshold between experimental and theoretical isotope intensities.
FeatureFinding (Mass Trace Detection)	mass_error_ppm	10.0	Mass error (ppm) for grouping ions into mass traces.
	min_trace_length	3.0	Minimum number of data points required for a mass trace.
	noise_threshold_int	1000.0	Intensity threshold below which signals are considered noise. Note: open mass raw data to select suitable threshold to be filtered out.
	chrom_peak_snr	6.0	Signal-to-noise ratio threshold for chromatographic peak detection.
FeatureFinding (Elution Peak Detection)	chrom_fwhm	3.0	Expected chromatographic peak width (full width at half maximum).
	min_fwhm / max_fwhm	1.0 / 60.0	Minimum and maximum allowed peak width.
FeatureFinding (Feature Detection)	local_mz_range	5.5	m/z range for grouping mass traces into features.
	mz_scoring_by_elements	true	Enables scoring by elemental composition (essential for DeepHalo).
	elements	"CHNOPSFCIBrINaFeSeB"	Elements considered for theoretical isotope pattern generation.
	remove_single_traces	true	Removes features consisting of only a single mass trace (essential for DeepHalo).
	isotope_filtering_model	"none"	Disables default isotope filtering, as DeepHalo performs its own classification.

For details on other parameters, see the config.toml file.